

# Boyce-Codd Normal Form (BCNF) Relational Database Design

> **Definition:** A relation is in **Boyce-Codd Normal Form (BCNF)** (also known as 3.5NF) if and only if for **every non-trivial functional dependency  $X \rightarrow Y$** , **X is a strict Super Key**.

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## 1. Detailed Technical Explanation

BCNF is a stricter version of 3NF that handles cases where a relation has **multiple overlapping candidate keys**.

### 3NF vs BCNF Comparison:

| Property | 3NF | BCNF |

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### Conversion Example to BCNF

#### Relation ADVISOR(Student\_ID, Subject, Advisor\_Name)

- **A**

#### Relation Status:

- Is it

#### BCNF Decomposition:

1. **AD**

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## 2. Core Concepts & Memory Keywords

- **Strict Super Key Rule:** Every determinant in a functional dependency must be a super key.

- **C**

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## 3. Must-Write Points for Exams

- BCNF eliminates all redundancy resulting from functional dependencies.

- Eve

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**4. Quick Recall Flow**

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3NF R